

Problem-Solution Fit

Date	06 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Small & Medium Scale Farmers • Agricultural Officers • Farm Owners	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited budget • Lack of technical knowledge • Poor internet connectivity • Limited access to smartphones	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Solutions • Agriculture expert advice • Weather apps • Government agriculture websites Pros • Easy to access • Basic farming guidance Cons • Not personalized • Information may not be accurate for every farm
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Wrong crop selection (Every season) • Unpredictable weather	9. PROBLEM ROOT / CAUSE [RC] • Lack of accurate crop recommendations	7. BEHAVIOR + ITS INTENSITY [BE] • Consults local farmers (High) • Checks weather forecast (Medium)

(Frequently) • Low crop yield (Every harvest) • High fertilizer cost (Often)	• Insufficient weather prediction • Poor soil analysis • Limited market information	• Uses traditional farming methods (High)
3. TRIGGERS TO ACT [TR] • Weather forecast changes • Soil condition reports • Crop disease alerts • Market price fluctuations	10. YOUR SOLUTION [SL] OptiCrop is an AI-powered smart farming system that: • Recommends the best crop based on soil and weather. • Uses a trained Machine Learning model for accurate crop prediction. • Displays crop details with images and Telugu crop names. • Improves productivity and crop selection accuracy. • Helps farmers increase crop yield and profit.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Mobile Apps • WhatsApp Groups • YouTube • Agriculture Websites OFFLINE • Agriculture Officers • Local Farmer Meetings • Seed & Fertilizer Shops
4. EMOTIONS BEFORE / AFTER [EM] Before • Confused • Worried After • Stressed • Confident • Satisfied • Hopeful		